WORKSHEET - NAPLAN Year 9

Measurement
- **Perimeter, Area and Volume**
- **Units of Measurement**

Geometry
- **Triangles and Other Geometric Properties**

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Exam Equivalent Time: 28.75 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Worked Solutions

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1. Measurement, NAP-26174

Convert each to centimetres

Kim $= 0.72 \times 100 = 72$ centimetres

Bob $= 815 \div 10 = 81.5$ centimetres

Liz $= 68$ centimetres

$\therefore$ Bob's hedge is the tallest

2. Measurement, NAP-08572

Since all sides of a rhombus are equal,

Perimeter $= 12$ cm

3. Measurement, NAP-32095

Consider the surface area of each pool:

1st pool $= 6 \times 19 = 114$ m^2

2nd pool $= 7 \times 18 = 126$ m^2

3rd pool $= 12.5 \times 12.5 = 156.25$ m^2

4th pool $= 10 \times 15 = 150$ m^2

$\therefore$ The 3rd pool, 12.5×12.5, has the largest surface area.

4. Measurement, NAP-32094

200 grams

5. Number, NAP-26177

Speed $= (390 \times 60)$

$= 23\,400$ metres per minute

6. Measurement, NAP-55611

mass

7. Algebra, NAP-79137

$S = ph + 2F$

8. Measurement, NAP-73221

The box front is around 4 cubes wide and 4 cubes high.

$\Rightarrow$ Estimated cubes that could fit

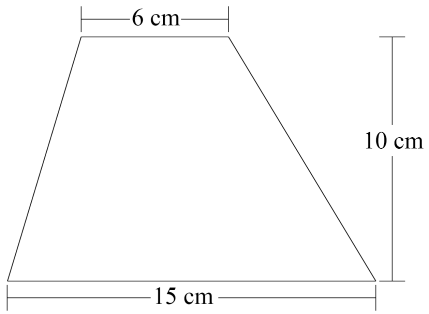
$$= h \times h \times (2h)$$
$$= 4 \times 4 \times (4 + 4)$$
$$= 16 \times 8$$
$$= 128$$

$\therefore$ 130 is the closest.

9. Geometry, NAP-32121

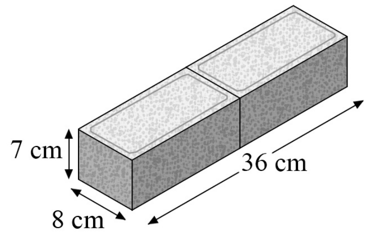
$$\text{Base length} = 8 \times 0.7$$
$$= 5.6 \text{ metres}$$

10. Measurement, NAP-49701



$$\text{Area} = \frac{1}{2}h(a + b)$$
$$= \frac{1}{2} \times 10 \times (6 + 15)$$
$$= 105 \text{ cm}^2$$

11. Measurement, NAP-38036



36 cm by 8 cm by 7 cm

12. Measurement, NAP-61561

$$C = 2\pi r$$
$$= \pi d$$
$$= \pi \times 3.2$$
$$= 10.05\ldots \text{ cm}$$

13. Measurement, NAP-08601

The semi-circles in both shapes have the same length
but have been turned around in Shape 2.

$\therefore$ They have different areas and the same perimeter.

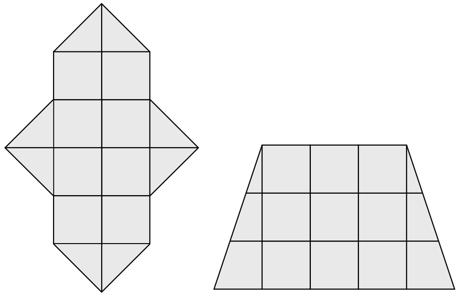
14. Measurement, NAP-96772

Let $d =$ length of other side

$$12.7 \times d \geq 60$$
$$d \geq \frac{6}{12.7}$$
$$d \geq 4.72\ldots \text{ m}$$

$\therefore$ Smallest length is 4.8 m (to 1 d.p.)

15. Measurement, NAP-72923



16. Measurement, NAP-08605

9 paving stones $= 18$ m²

$\Rightarrow$ 1 grid square $= 2$ m²

$\therefore$ Total grass $= 27$ grid squares

$= 27 \times 2$

$= 54$ m²

17. Measurement, NAP-26209

1.46 metres

18. Measurement, NAP-02688

$$C = 2\pi r$$
$$90 = 2 \times \pi \times r$$
$$r = \frac{90}{2\pi}$$
$$= 14.32\ldots \text{ cm}$$

19. Measurement, NAP-96773

Diameter $= 530$ mm

$= 0.530$ m

$\therefore$ Distance (12 circumferences)

$= 12 \times (\pi \times d)$

$= 12 \times (\pi \times 0.530)$

$= 19.98\ldots$ m

$= 20$ m (nearest m)

20. Measurement, NAP-26237

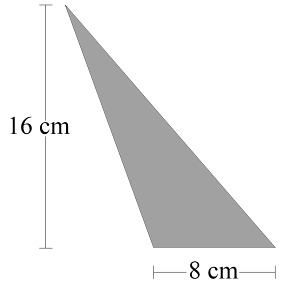
Metres of door casing needed

$= (2110 \times 2) + 2520$

$= 6740$ mm

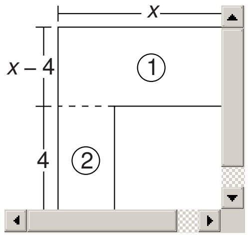
$= 6.74$ m

21. Measurement, NAP-96775



$$\text{Area} = \frac{1}{2}bh$$
$$= \frac{1}{2} \times 8 \times 16$$
$$= 64$$

22. Algebra, NAP-08636



$$\begin{aligned}\text{Area} &= \text{Area 1} + \text{Area 2}\\ &= x(x-4) + 4(x-5)\\ &= x^2 - 4x + 4x - 20\\ &= x^2 - 20\end{aligned}$$

23. Measurement, NAP-38098

$1222 \text{ mm} = 1.222 \text{ m}$

Since the area is 1m^2,

$$1.222 \times \text{Width} = 1$$

$$\begin{aligned}\text{Width} &= \frac{1}{1.222}\\ &= 0.8183\ldots\end{aligned}$$

$$\therefore\ \text{Width} = 818 \text{ mm} \ \ (\text{nearest mm})$$